An AI-ready evidence base on product testing with users in agriculture

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Abstract

Breeding programs decide each season which varieties to advance, and those decisions increasingly depend on whether a variety performs well in farmers' fields, meets the demand of farmers, processors and consumers, and copes with climate risk. Since it was first proposed in 2011, the tricot approach to decentralized on-farm testing has produced a large body of evidence on exactly these questions (van Etten, 2011; de Sousa et al., 2024), but it is spread across journal articles, guides, reports, briefs and presentations that breeders rarely have time to search. AI assistants could close this gap, yet an assistant without curated sources may invent results, misreport numbers or stretch a finding from one crop and country into a general rule. We present an AI-ready evidence base that distils fifteen years of openly available publications on tricot and related approaches to product testing with farmers and consumers into verified markdown cards, each recording the questions a source answers, its methods and stated limitations, key numbers with their primary sources, and links to the related data and code. The repository works as a skill for any AI assistant, with explicit rules for evidence boundaries, attribution and contact consent. It is the evidence layer of a set of linked AI-ready resources, developed within the CGIAR Digital Transformation Accelerator, on published evidence, field protocols, on-farm trial data and analytical approaches for demand-led variety advancement.

Keywords: demand-led breeding, variety advancement, tricot, on-farm testing, citizen science, knowledge base, AI-ready documentation, FAIR principles

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The most recent version of the knowledge base and this paper is always available at the repository link above.

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Statement of need

Late-stage variety testing is where breeding programs decide which candidates move toward release. Those decisions now ask more than whether a variety yields well on station. Breeders need evidence that a variety performs in the environments where farmers grow it, matches what farmers, processors and consumers value, and holds up under the climate risks of the target region. The tricot approach was designed to produce this evidence. First proposed as a way to crowdsource crop improvement with farmers (van Etten, 2011), it asks farmers to test three technologies out of a larger set, assigned at random, and to rank them in their own fields (van Etten et al., 2019). Over fifteen years, the approach has been applied to more than 30 crops in more than 25 countries, and extended to consumer testing and product concept testing (de Sousa et al., 2024). Combined with genomic, socio-economic and climate data, tricot rankings have improved the prediction of on-farm performance (de Sousa et al., 2021) and revealed farmer segments with distinct variety preferences (Voss et al., 2025).

This evidence answers questions breeders face every season. How many farms does a late-stage trial need? Do farmer rankings agree with measured yield, and will a release committee accept them? Which traits drive preference, and do women and men, or different market segments, prefer different varieties? Yet the answers sit in dozens of documents of different types, written for readers who already know where to look.

AI assistants can bring this evidence to breeders in the moment they need it, but only if they answer from verified sources. An assistant working from general training data can invent a paper, quote a number from the wrong study, or present a result from one crop and country as a general rule. When findings feed into advancement and release decisions, these errors carry real costs.

We built the knowledge base so that breeders, technicians and students, and the AI assistants they work with, can consult this evidence the way they would consult a colleague who knows the work and the people behind it.

Place in the AI-ready suite

The knowledge base is the evidence layer of ai-ready-documentation (`https://github.com/AgrDataSci/ai-ready-documentation`), a set of linked repositories in which each layer carries its own skill. The other layers hold the field protocols, the harmonized on-farm trial data and the team's analytical approaches, and a planned layer will link the trial data with climate, soil and other environmental data. A shared template supports reproducible analyses across them (de Sousa and Laporte, 2026). The suite and its design are described in the ai-ready-documentation repository.

The layers connect through shared identifiers. A card names the dataset its findings rest on, a dataset names the protocol that produced it, and protocols and data share trait definitions from the Crop Ontology. An assistant can follow these links from a published finding to the trials behind it and, once the integration layer is in place, to the climate data that help explain it.

Content

The knowledge base holds one card for each source document, covering fifteen years of published evidence, from the first proposal of the approach (van Etten, 2011) to the current tricot handbook (Chase et al., 2026). Most sources concern tricot, but the knowledge base also covers related work on product testing with users, such as consumer testing of processed foods, product concept testing with videos, and data standards for breeding. Most sources are peer-reviewed research articles. The others include software papers, reviews, guides, reports, policy briefs, working papers, preprints and presentations, each labelled by type so that an assistant can weigh it accordingly. The repository grows as new work is published, and each change updates the version of the skill. The cards cover pre-breeding with genebanks, the origins and guides of the approach, data quality and impact, climate and genomic analysis, software and data standards, applications to specific crops, consumer and product concept testing, gender and inclusion, and the mainstreaming of tricot in breeding programs.

Each card follows one template. Its metadata give the DOI or repository handle, a stable open copy, the license, the dataset and code behind the paper, the crops and regions covered, search keywords and a one-sentence gist. Its body lists the questions the source answers, a summary of 400 to 500 words written in our own words, the methods in brief, the limitations stated by the authors, a table of key numbers with their primary sources, and notes on discrepancies with other cards. An optional field records claims the source is likely to be stretched to but does not support.

Repository structure

```
tricot-knowledge-base/
|-- SKILL.md            # Skill instructions, version and paper index
|-- people.md           # People behind the papers, naming and consent
|-- AGENTS.md           # Entry point for AI agents
|-- llms.txt            # Short map of the repository for AI tools
|-- index.json          # Version, card hashes and links (generated)
|-- relations.jsonld    # Links between cards as JSON-LD (generated)
|-- papers/             # One card per source
|-- papers-pdf/         # Source documents and their licenses
|-- TEMPLATE.md         # Template for new cards
|-- scripts/            # Rebuilds the index and version
|-- tests/              # Evaluations of card format and answers
|-- docs/               # This paper
|-- CITATION.cff
|-- LICENSE
```

Key features

- **Verified numbers.** Every number in a card is checked against the full text of its source. Disagreements between a paper's text and tables, or between papers, are recorded in the card notes rather than smoothed over.
- **Primary sources.** When a paper reports a figure from another study, the card names that study, so users cite the original.
- **Linked papers.** Cards declare how they build on each other with terms from the Citation Typing Ontology (Peroni and Shotton, 2012), such as `extends`, `usesDataFrom` and `disagreesWith`. A script checks every link, adds the inverse links and exports the graph as JSON-LD, so an assistant can follow a line of work from one paper to the next.
- **Evidence boundaries.** The skill asks the assistant to separate direct evidence, indirect evidence, methodological inference and evidence gaps. It must keep each study's crop, country, sample and design when it transfers a finding, and never present a study-specific sample size as a universal rule.
- **People and consent.** A people file records how to credit the researchers behind each card and whether they agreed to be suggested as contacts. The assistant credits work by name but never speaks for a person, suggests a contact only with recorded consent and through an institutional channel, and never shares personal contact details or photos.
- **Versioning and updates.** A script computes a version from the content of every card and writes it to the index. A GitHub Action rebuilds the index and publishes the packaged skill on every change. Once per conversation, an installed skill compares its version with the live repository and fetches only what changed.
- **Light skill.** The assistant loads a short index first and opens only the cards a question needs, usually two to five. The packaged skill stays small, so it travels easily between users.
- **Open and readable.** Cards are plain markdown under a CC BY 4.0 license. Source documents keep their original licenses, which are listed for every file.

Verification and evaluation

We tested two card formats on the same source. A short card held a summary and key numbers. A long card added the questions the paper answers, the methods and the stated limitations. Answering expert questions from one card at a time, the long format gave more complete and better-grounded answers, for example the reasoning behind a recommended trial size rather than the number alone. All cards now use the long format.

A second evaluation probes whether an assistant keeps evidence boundaries, regulatory distinctions, primary-source provenance and contact consent. It poses adversarial questions, such as whether tricot is approved for releasing a crop in a country without a primary record, or whether a researcher's email can be found online when consent is pending. Answers are scored on factual accuracy, provenance, scope, uncertainty and consent. Any fabricated source, unsupported claim of regulatory approval or breach of consent counts as an automatic failure. The evaluation is meant to be rerun after every change to the skill.

Supporting variety advancement decisions

The knowledge base answers the evidence questions that come before and after an advancement decision. A breeder planning a late-stage trial can ask how many farmers and entries are needed, how to combine farmer rankings with yield measured on a subset of plots, or how to sample participants so that women and different user segments are represented. A breeder reading results can ask how to interpret rankings across environments, how climate shapes variety performance, or how concept testing has been used to define product profiles. Each answer comes from verified cards, cites the primary source, keeps the context of the original study, and says where the evidence stops.

Through the data layer, the same skills let an assistant move from published evidence to the trial data of a specific crop and region, and the planned integration layer will add agroclimatic information. Together, these resources form the foundation for AI-supported recommendations that match candidate varieties with farmer and market demand and with climate risk.

The rules in the skill also serve as practical guidance for responsible use. They separate measured yield from rankings, candidate advancement from official release, and direct evidence from extrapolation. They also protect the people behind the work. The same cards can support the training of breeders and technicians in demand-led on-farm testing.

Implementation and usage

Users reach the knowledge base in four ways.

1. **In Claude.** Download the latest skill package from the repository releases and upload it as a skill. The assistant then consults the index whenever a question touches tricot or on-farm testing.
2. **In Claude Code.** Clone the repository into the skills directory.
3. **By sharing the link.** Paste the repository link into an AI assistant with web access. It reads the agent instructions and the skill, answers from the live files, and offers to install the skill.
4. **By hand or with other tools.** Start from the index table in `SKILL.md` and open the relevant cards in `papers/`.

Contributors add a source by extracting its text, drafting a card from the template, checking every number against the source, adding links to related cards and authors, and opening a pull request. The index and version update automatically when the change is merged.

Limitations

The knowledge base covers openly available sources produced by or with the tricot team, so it reflects the perspective of the people who developed the approach. It is a curated synthesis, not a systematic review, and the absence of a topic means it is absent from these sources, not from the literature. Summaries are written by the team and checked against the sources, but they remain interpretations, and users should cite and consult the original papers. The knowledge base informs advancement decisions but does not make them. Recommendations for specific candidates require the trial data, environmental data and analyses of the other layers, several of which are still under development. Answers depend on the assistant following the rules in the skill, which the evaluations test but cannot guarantee. Consent to be suggested as a contact is collected person by person, so the assistant suggests contacts only for those who have agreed.

Conclusion

Breeders need evidence on farmer and market demand and on climate risk at the moment they decide which varieties to advance. Curated, verified and openly licensed evidence lets AI assistants deliver it accurately, credit the people behind the work and say where the evidence stops. The knowledge base grows with the literature, stays current in every installed copy, and connects published findings to the data and analyses that support demand-led breeding.

Code and data availability

The knowledge base, its source documents and this paper are openly available on GitHub at `https://github.com/AgrDataSci/tricot-knowledge-base`. The cards and documentation are released under a CC BY 4.0 license. Source documents keep their original licenses, listed in `papers-pdf/LICENSES.md`.

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